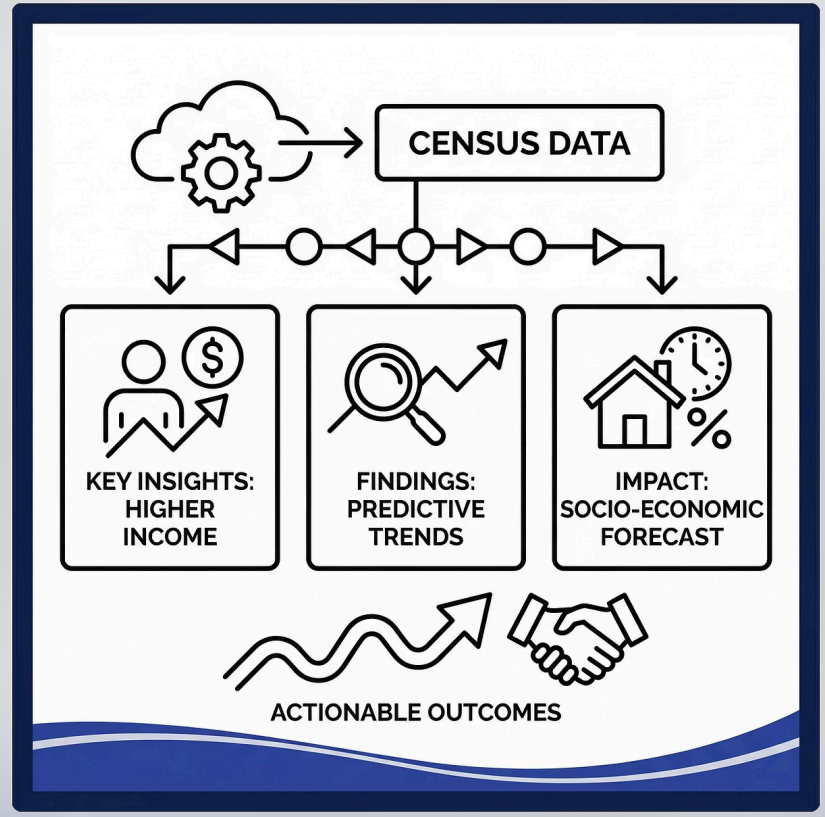
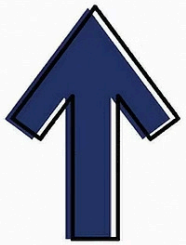
Predicting Adult Income Using Census Data

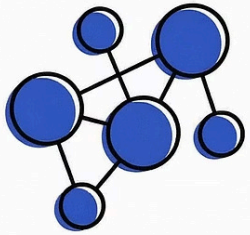
A Complete Data Mining Workflow

By: Khaydie Ramos

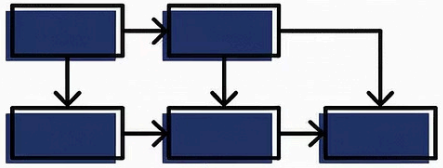




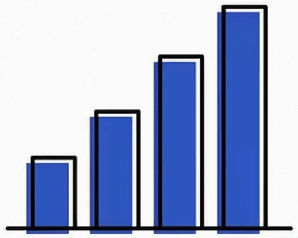
OBJECTIVE



DATASET



PIPELINE



EVALUATION METRICS

Project Overview

Objective

Predict whether individual income exceeds $50K annually using demographic and employment data

Dataset

UCI Adult Census dataset with comprehensive demographic information

Pipeline

Data Cleaning → EDA → Feature Engineering → Selection → Modeling → Tuning → Ensemble

Evaluation

Accuracy, Precision, Recall, and F1-Score metrics

Motivation & Importance

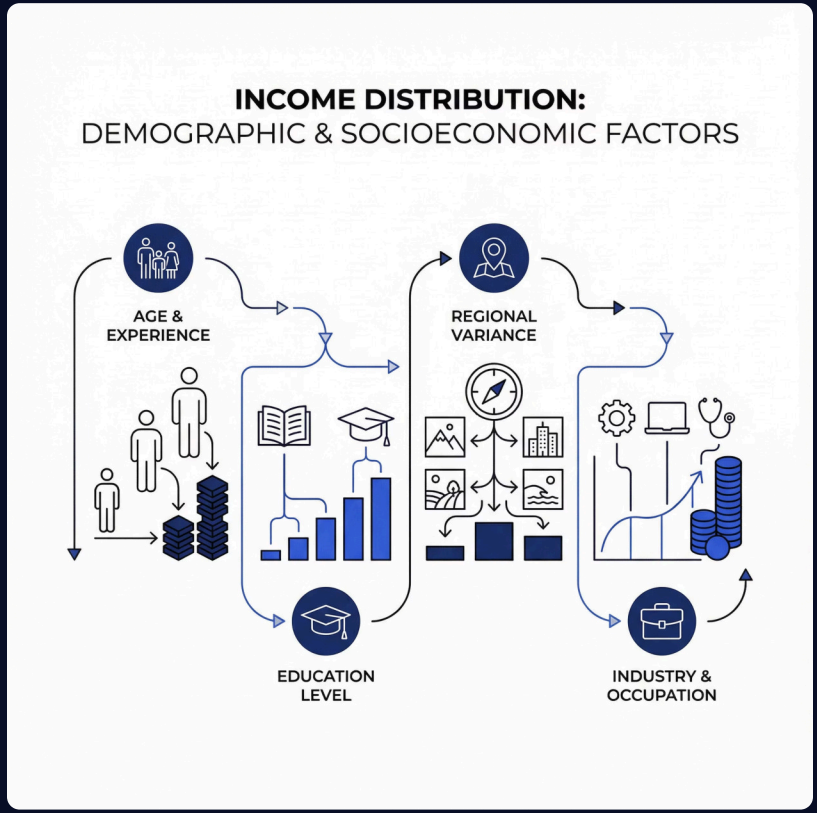
Real-World Impact

Income prediction models have significant applications across multiple domains:

- **Public Policy:** Informing economic programs and social services allocation
- **HR Analytics:** Salary benchmarking and compensation planning
- **Economic Modeling:** Understanding factors that drive income inequality

Dataset Characteristics

The Adult Census dataset provides a rich combination of categorical and numeric variables, making it an excellent benchmark for classification techniques and feature engineering approaches.



Dataset Summary

1

Size & Structure

Approximately 48,000 rows with 15 features capturing diverse demographic information

2

Feature Categories

Demographics (age, race, sex), employment details (occupation, workclass), and financial indicators (capital-gain, capital-loss)

3

Target Variable

Binary classification: income ≤$50K vs income >$50K

4

Class Distribution

Notable class imbalance with approximately 24% of samples in the >$50K category

Initial Data Quality Issues

Comprehensive data quality assessment revealed multiple challenges requiring systematic treatment:

Missing Values

Significant gaps in workclass, occupation, and native-country fields requiring imputation strategies

Outliers

Extreme values detected in capital-gain, capital-loss, hours-per-week, and age variables

Skewed Distributions

Heavy right-skew in capital-gain and capital-loss affecting model performance

Redundancy

Education and education-num contain duplicate information

Sparse Categories

Native-country feature contains numerous rare values requiring grouping

Outlier Detection & Treatment

IQR-Based Approach

Applied Interquartile Range (IQR) method for systematic outlier identification across numeric features.

Capital Variables

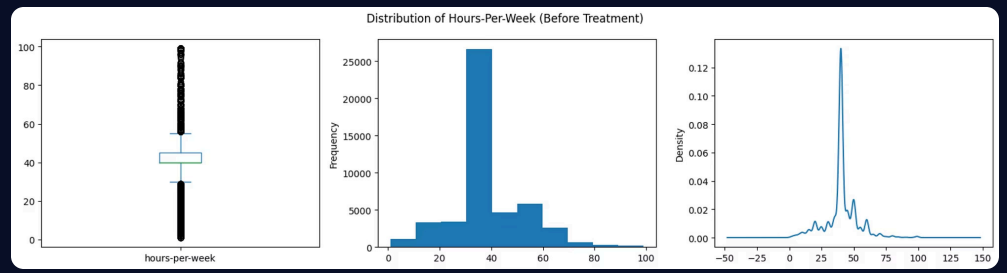
- Capped extreme capital-gain and capital-loss values
- Preserved information while reducing variance

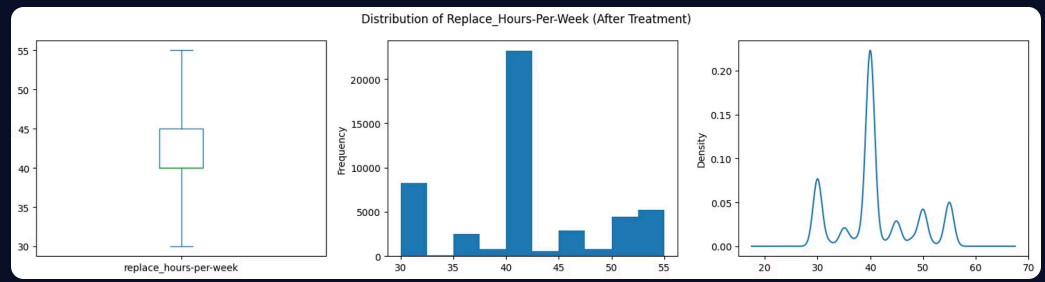
Hours Per Week

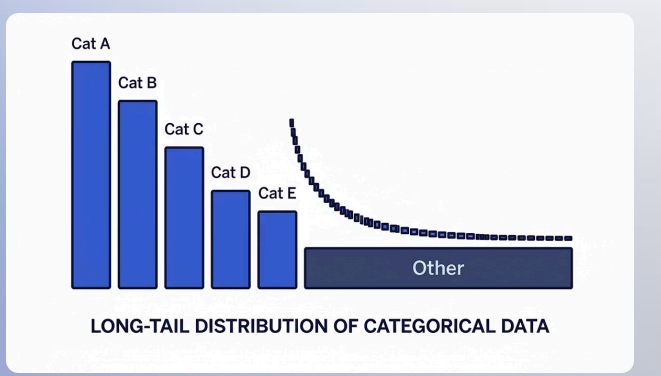
- Set floor at 30 hours (part-time threshold)
- Set ceiling at 55 hours (reasonable maximum)

Age Treatment

Replaced outlier age values with statistically reasonable bounds to maintain demographic validity.







Categorical Cleaning

01

Rare Value Consolidation

Grouped native-country values appearing in less than 1% of samples into an "Other" category, reducing dimensionality while preserving information

02

Label Standardization

Cleaned and standardized text labels across all categorical features to ensure consistency

03

Redundancy Removal

Dropped the education column after confirming education-num captures the same ordinal information

04

Encoding Preparation

Prepared categorical variables for one-hot encoding to enable machine learning compatibility

Feature Engineering

Strategic feature creation to capture non-linear relationships and domain knowledge:

Age Binning

Created meaningful life stages:

- Young Adult (<25)
- Mid-Career (25-35)
- Senior (35-50)
- Retirement (50+)

Binary Indicators

New flags to capture key patterns:

- **has_gain:** Capital gains present
- **has_loss:** Capital losses present
- **works_long_hours:** >50 hrs/week

Preprocessing

Final transformations:

- One-hot encoding for categorical variables
- Standard scaling for continuous features



Feature Selection with SelectKBest

Applied Mutual Information criterion to identify the most predictive features for income classification.



95%

Age

Strongest predictor of income level



88%

Education-num

Years of education highly correlated



76%

Hours-per-week

Work commitment indicator



72%

Marital Status

Family structure impacts income



68%

Occupation

Job type determines earnings

Modeling Strategy



Baseline Models

K-Nearest Neighbors, Naive Bayes, and Decision Tree algorithms trained on cleaned features



Performance Comparison

Evaluated baseline models using multiple metrics to identify top performers



Hyperparameter Tuning

Grid search optimization for best-performing models



Ensemble Method

Soft voting ensemble combining multiple algorithms

Baseline Model Performance

Initial model training revealed significant performance differences across algorithms:

K-Nearest Neighbors

Accuracy: ~0.82

Strongest baseline performer, capturing local patterns effectively

Decision Tree

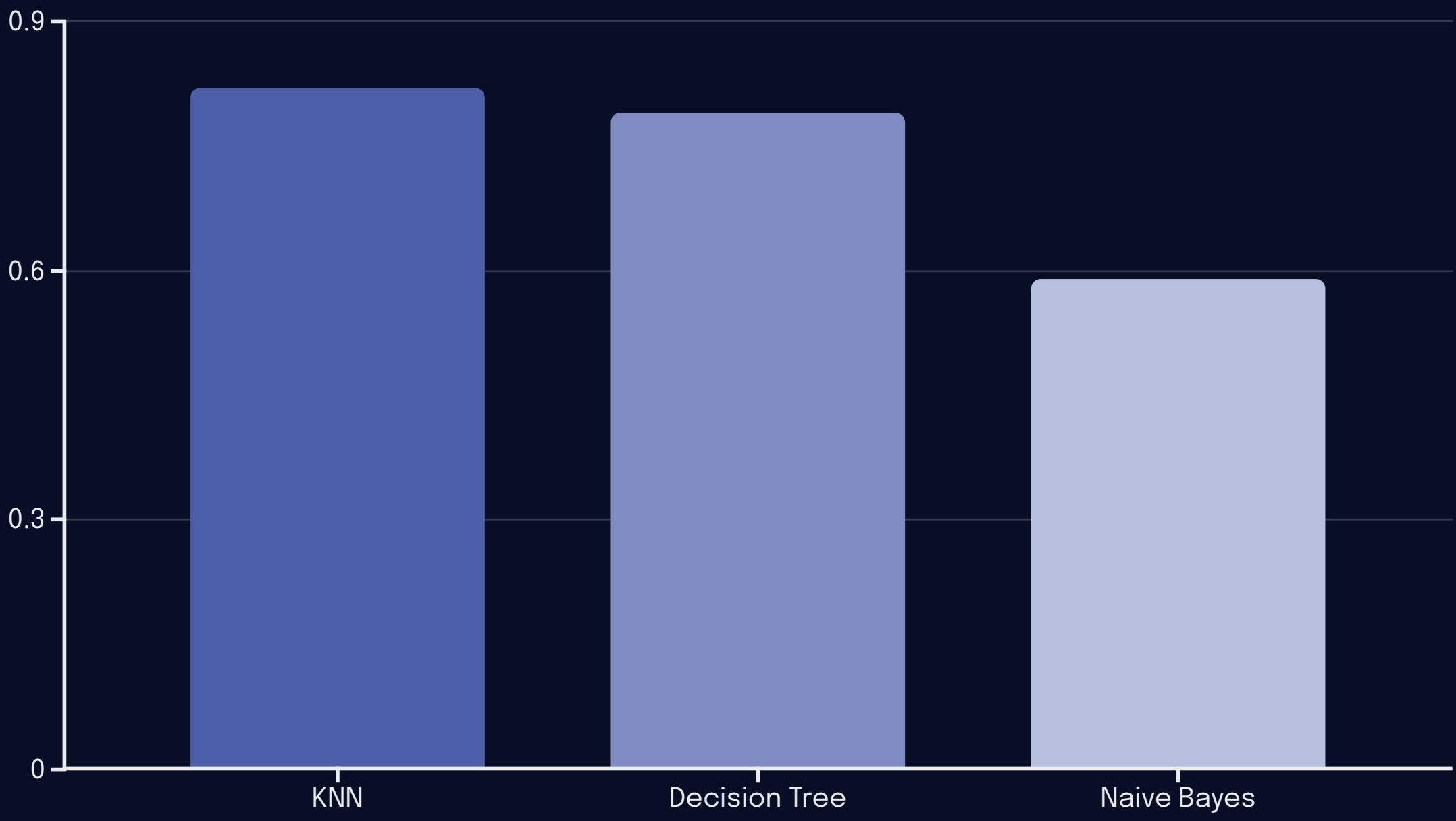
Accuracy: ~0.79

Solid performance with interpretable decision rules

Naive Bayes

Accuracy: ~0.59

Struggled with feature dependencies and violated independence assumptions



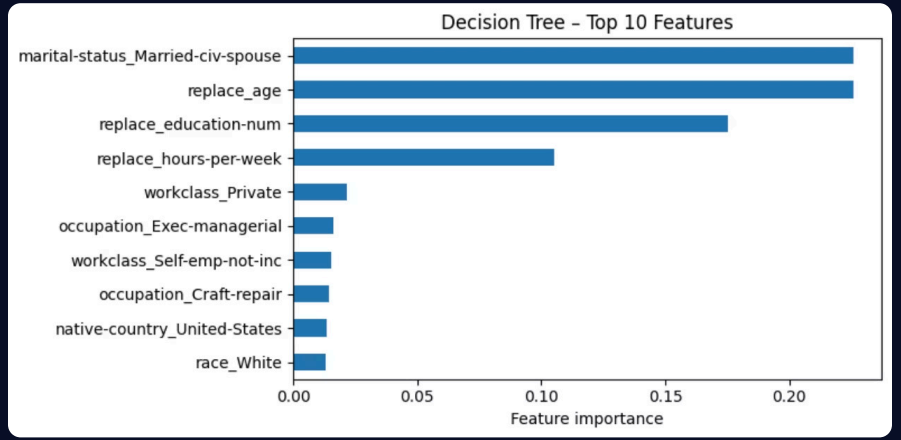
Feature Importance Insights

Decision Tree Analysis

Feature importance scores revealed marital status as the most predictive factor, followed by education level and age.

Top Predictors

1. **Marital Status:** Married individuals show higher income probability
2. **Education-num:** Strong linear relationship with income
3. **Age:** Experience and career progression indicator
4. **Hours-per-week:** Work commitment correlation
5. **Occupation & Relationship:** Job type and household structure



Permutation importance analysis confirmed these findings, validating the feature selection strategy.

Reduced Feature Set Modeling

1

Feature Reduction

Began with 100+ features and selected top 21 features based on mutual information scores and domain importance

2

Improved Stability

Reduced dimensionality led to more stable model predictions and better generalization

3

Accuracy Gains

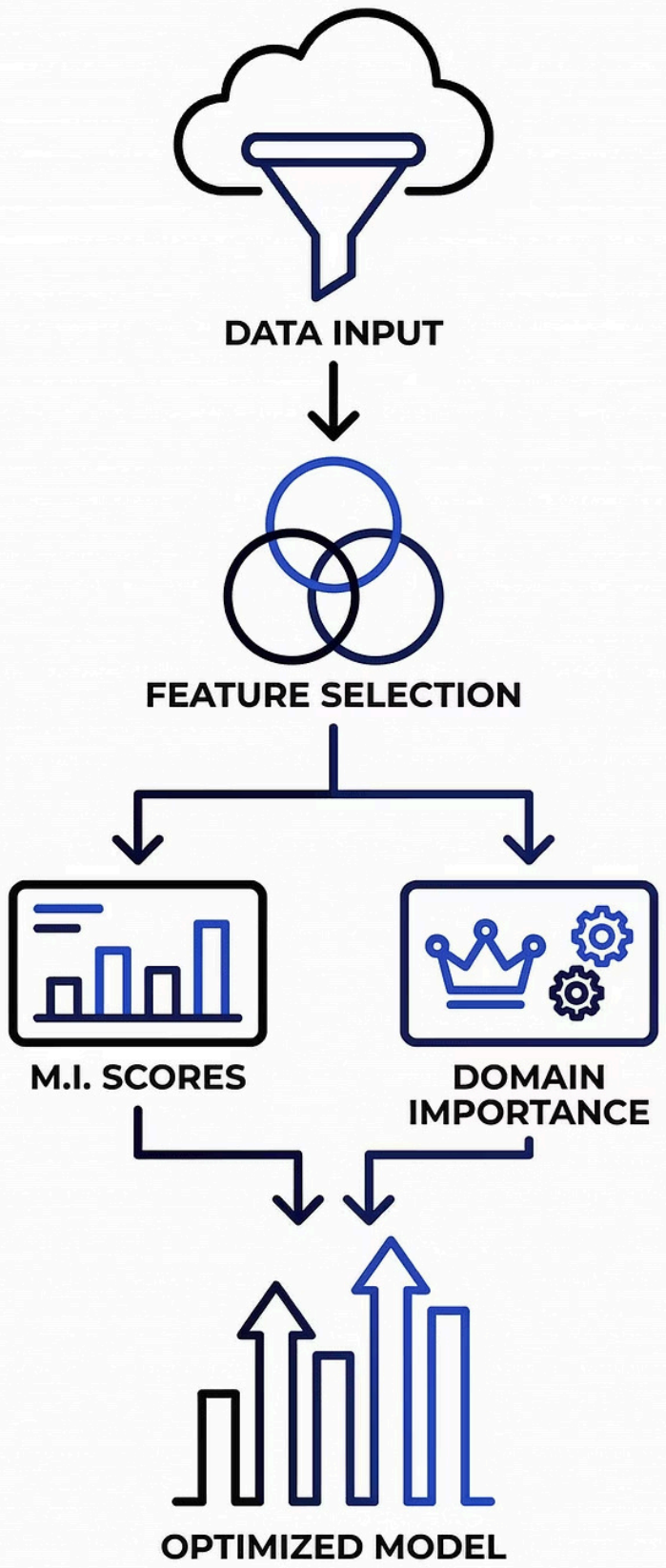
Slight improvements in accuracy across models by reducing noise from low-importance features

4

Training Efficiency

Significantly faster training times with maintained or improved performance

Selected 21 Features Include: age, education-num, hours-per-week, capital-gain, capital-loss, marital-status categories, occupation categories, relationship types, and engineered binary flags.

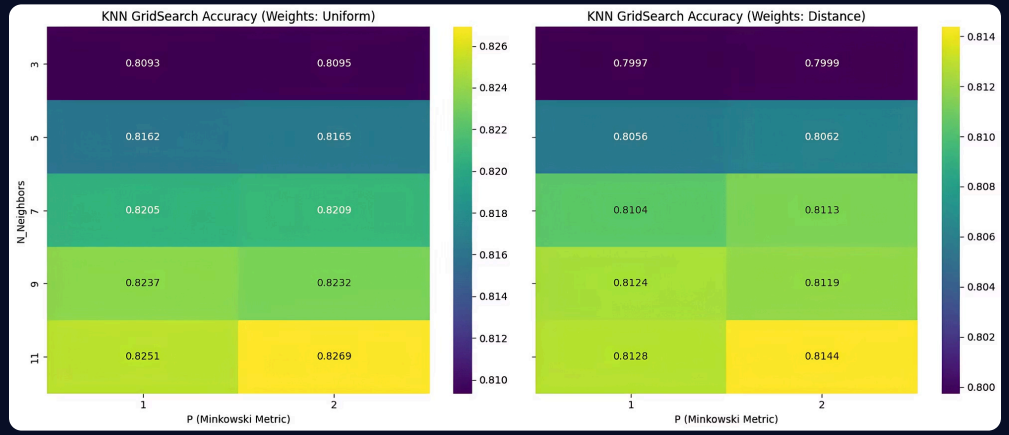


Hyperparameter Tuning

K-Nearest Neighbors

Grid search identified optimal configuration:

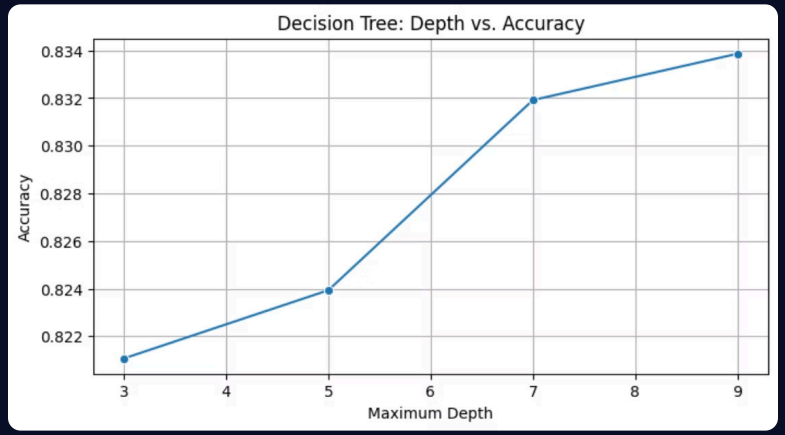
- **k = 11:** Balance between bias and variance
- **p = 2:** Euclidean distance metric
- **Weights = uniform:** Equal neighbor contribution



Decision Tree

Best parameters discovered through systematic search:

- **max_depth = 9:** Optimal tree complexity
- **criterion = entropy:** Information gain splitting
- **min_samples_leaf = 10:** Prevents overfitting



Final Tuned Model Performance

Hyperparameter optimization yielded significant improvements over baseline models:

83.4%

Decision Tree

Best overall accuracy after tuning

82.8%

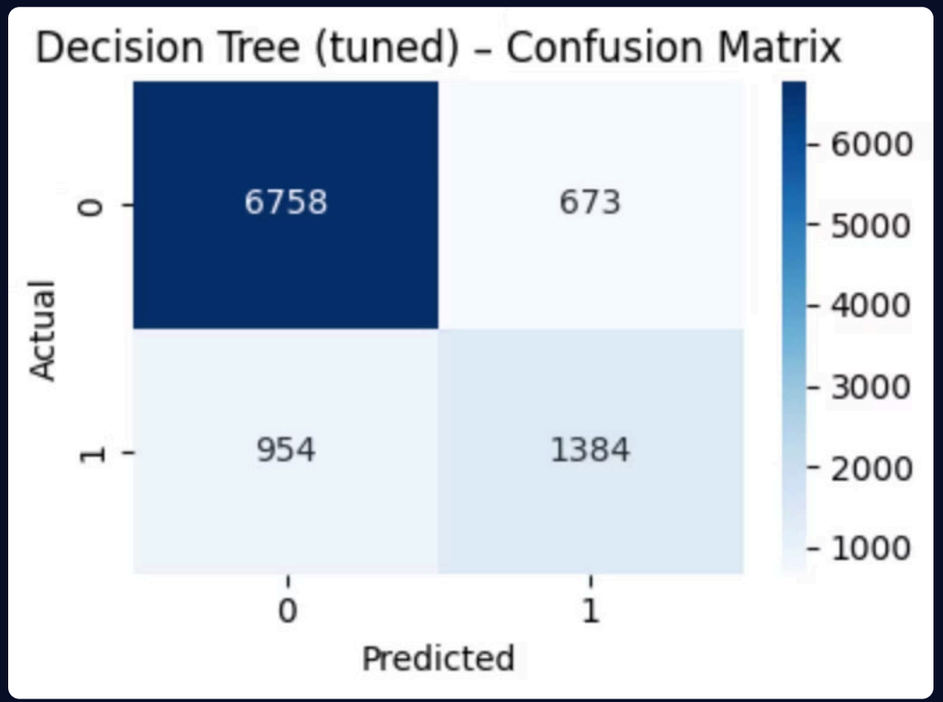
K-Nearest Neighbors

Strong performance with optimized parameters

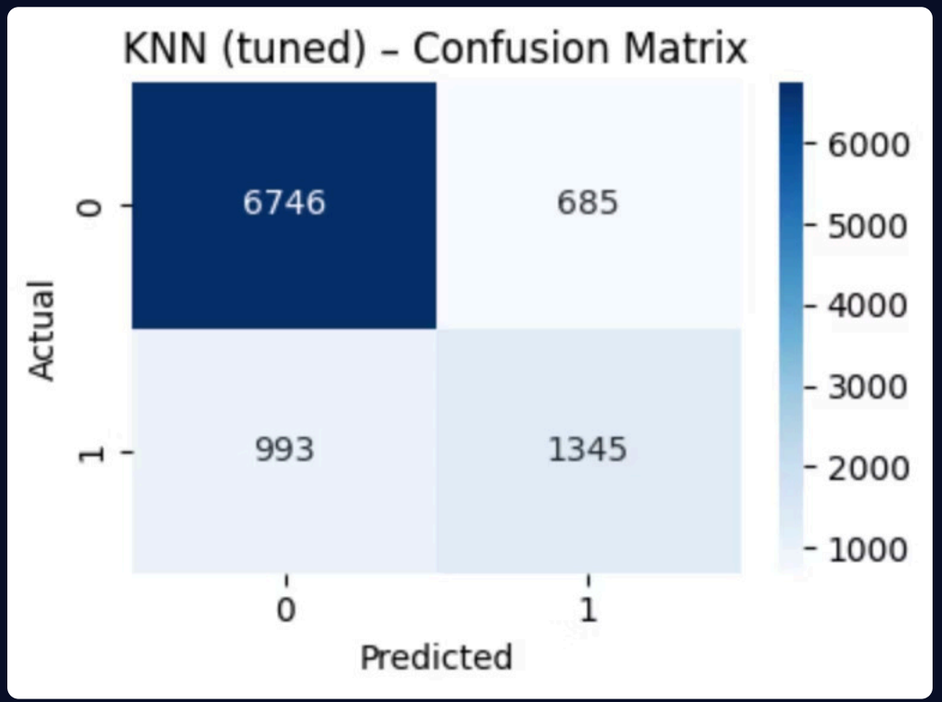
+4.4%

Improvement

Average gain over baseline models



Tuned Decision Tree: Balanced precision and recall with strong true positive rate



Tuned KNN: Competitive performance with slightly higher false positive rate

Ensemble Method Results

Decision Tree

Accuracy: 0.834



K-Nearest Neighbors

Accuracy: 0.828

Naive Bayes

Accuracy: 0.59

Soft Voting Ensemble: ~0.80 Accuracy

The ensemble underperformed the best individual model (Decision Tree at 0.834). This occurred due to insufficient error diversity among the base learners - the models made similar mistakes, limiting the ensemble's ability to correct individual errors.

Key Insight: Ensembles require diverse models with complementary strengths. When one weak model (Naive Bayes) drags down the average, soft voting may not improve performance.

Key Takeaways

Data Quality Matters Most

The largest performance improvements came from thorough data cleaning and strategic feature engineering, not from complex algorithms

Critical Predictors Identified

Education level, age, marital status, and hours worked emerged as the strongest income predictors

Best Model: Tuned Decision Tree

Achieved 83.4% accuracy with excellent interpretability and balanced precision-recall trade-off

Ensemble Limitations

Soft voting ensembles don't always improve performance – they require diverse, complementary base models

Limitations & Future Work

Current Limitations

Class Imbalance

Only 24% of samples are >$50K, potentially biasing predictions toward the majority class

Limited Context

Missing variables like industry, location details, and benefits that influence income

Algorithm Limitations

Naive Bayes proved unsuitable due to feature dependencies

Future Directions



Advanced Algorithms

Test Random Forest and XGBoost for improved performance



SMOTE Oversampling

Address class imbalance with synthetic minority samples



Interaction Features

Engineer features capturing relationships between variables



K-Fold Cross-Validation

Implement stratified k-fold for robust evaluation



Questions?

Thank you for your attention!